AWS & FORBES INSIGHTS

Enterprise priorities for workload migration and modernization

A global enterprise study of strategies and goals for Microsoft, Oracle, SAP, VMware, and mainframe workloads



INTRODUCTION

Migrating enterprise workloads to drive transformation and business impact

Digital transformation isn't just about implementing new technology; it's also about using technology to achieve business goals faster. Reinventing your business often begins by taking steps to invest in shifting strategies, processes, people, and culture. In the current business climate, you are likely being asked to drive that change in the face of economic uncertainty, rising costs, and even reduced staffing. It's important to remember that transformation is neither about scope nor size—it's about meaningful impact.

Quite often, the systems that are central to your strategy and process are the mission-critical applications that your business relies on every day— from companies such as Microsoft, Oracle, SAP, VMware and other established software vendors. These systems and the data associated with them are at the heart of any digital transformation you're likely to undertake. They are centrally tied to the products and experiences that you deliver and the decisions that you make every day.

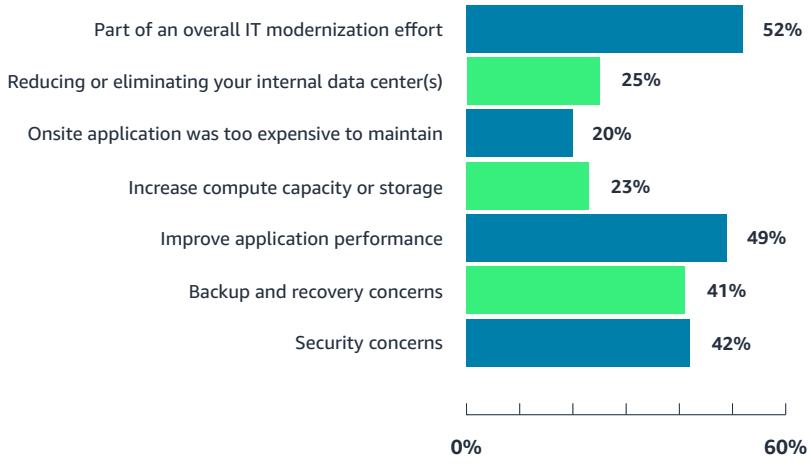
Any legacy systems can be costly in terms of licensing, infrastructure, maintenance, and staffing, and they are often difficult to scale and modernize within the constraints of your own data centers.

> **"94% of enterprises plan to increase cloud spending in 2023."**
>
> *Forbes*, 2023

How much is keeping mission-critical applications and data on premises holding you back from transforming your business?

A 2023 *Forbes* survey of 1,000 organizations found strong support for migrating and modernizing enterprise workloads. Ninety-four percent of the survey respondents identified plans to increase cloud spending this year, with a particular focus on funding modernization and transformation initiatives, reducing application maintenance costs, replacing and scaling aging infrastructure, and increasing performance and security.

Reasons for migrating enterprise workloads to the cloud



ENTERPRISE RESEARCH

Migrating and modernizing core business applications

The *Forbes* survey evaluated current migration objectives, readiness, goals, and strategies of organizations and their leaders to move core business applications from on-premises systems to modern cloud platforms. The survey assessed a range of key issues, including:

- The current operation of enterprises relative to their mission-critical workloads
- The role of the modern cloud in achieving current and future objectives
- Perceived advantages and expected outcomes of migration
- Potential obstacles standing in the way of successful modernization
- Realization of financial and productivity benefits from these platforms

The sampling of data on the following pages is provided to help you understand where other enterprises have progressed in their cloud journeys and consider some of the business drivers behind their migration and modernization strategies. Survey details are organized by the primary enterprise workload types that Amazon Web Services (AWS) is helping customers modernize for today and tomorrow. They also highlight the available programs, services, and partners that you can engage with to propel your business transformation.

AWS for Microsoft workloads

For decades, businesses have chosen Windows Server and SQL Server as the foundation for their mission-critical workloads. As they embrace digital transformation and develop their plans to move these workloads to the cloud to drive innovation and agility, many of these businesses reach a decision point on what to do with them. Sometimes such workloads are best migrated “as is,” lifting and shifting them to equivalent or newer versions in the cloud. In other cases, they are better off being replaced by open-source or specialized cloud operating systems and databases.

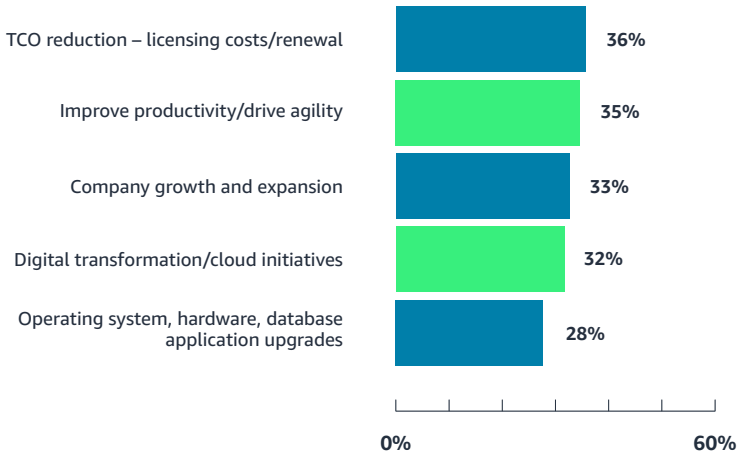
To complicate this decision, customers may have to deal with Microsoft end-of-support (EOS) events and annual license renewals. As you evaluate cloud service providers, determine if they can meet the needs of your applications and business by providing the ability to lift and shift or refactor when appropriate and offering complementary services to drive new cloud experiences for your employees and customers.



Key findings: Microsoft workloads

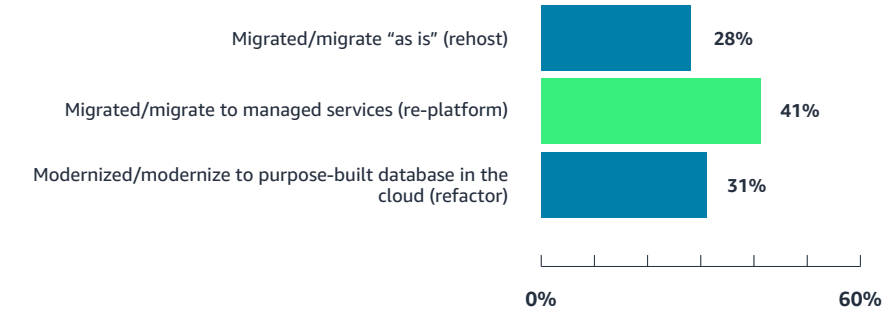
The top drivers for enterprises to migrate Windows workloads are lower costs and greater agility. It is interesting to note that these drivers exist in addition to the need for digital transformation. This means that a growing number of today's enterprises may be specifically focusing on Windows Server migration and modernization outside of an overall modernization effort.

What are your top drivers for migrating **Microsoft Windows Server** applications to the cloud?

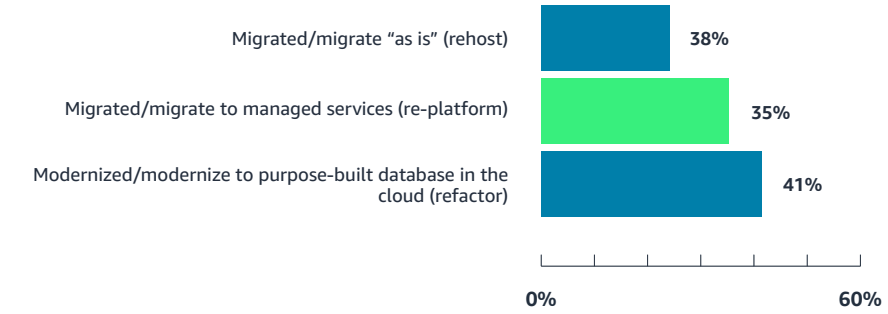


Enterprises are more likely to move to managed services or choose purpose-built cloud offerings over simply rehosting existing vendor offerings. With a major EOS event for SQL Server in 2022 and an upcoming EOS for Windows Server in 2023, this indicates that enterprises are evaluating alternatives to traditional software vendors and licensing models.

What type of migration did you pursue or are you planning for **SQL Server** in the cloud?



What type of migration did you pursue or are you planning for **Microsoft Windows Server** applications in the cloud?



AWS FOR MICROSOFT

Get started with AWS for Microsoft workloads

AWS offers proven migration and modernization services and programs to help you design a transformation strategy for your Microsoft workloads. If you are planning to migrate and modernize your applications and data, contact AWS or one of our **Microsoft Workloads Competency Partners**. Engage an experienced AWS Microsoft Specialist for a free **AWS Optimization and Licensing Assessment** (AWS OLA) today for insights into how AWS can help you save on third-party licensing costs and run your resources more efficiently. Or, explore our **Migration Acceleration Program (AWS MAP) for Windows**, which provides tools to help you reduce costs while automating and accelerating execution.



AWS for SAP

SAP is the leading provider of enterprise resource planning (ERP) systems that help customers run core business processes for departments such as finance, manufacturing, HR, supply chain, services, and procurement. By providing a unified system of record for information across the enterprise, SAP aims to help customers run their businesses with greater operational visibility and transparency.

As cloud computing has matured and customers' needs for their ERP systems have evolved, the cloud has quickly become the preferred platform to drive new and faster business insights, increase scale and performance, and accelerate the delivery of business-critical data enterprise-wide.

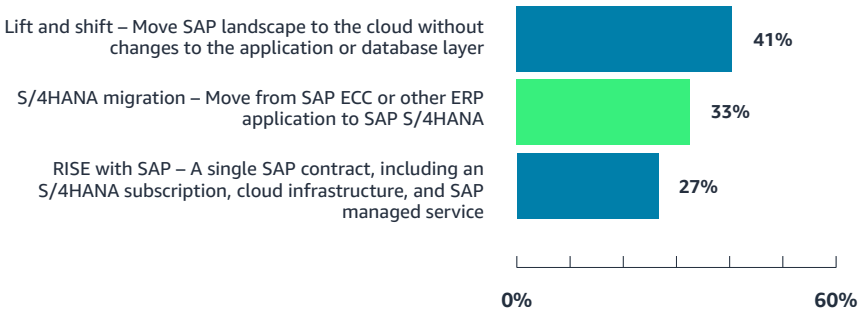
Teams exploring a digital transformation strategy should consider SAP applications a key component of their cloud migration and modernization strategy.



Key findings: SAP workloads

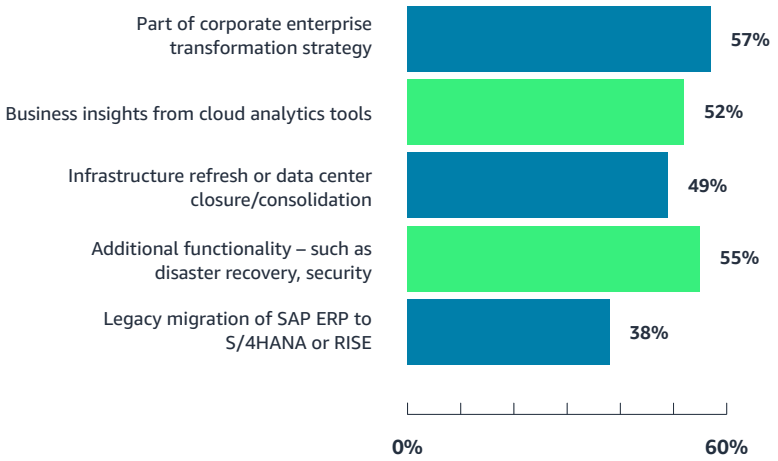
Enterprises are mixed on how they want to migrate SAP to the cloud, with lift and shift (rehosting of existing SAP applications and databases) being the preferred method.

Which cloud option best defines your SAP migration strategy?



SAP migration and modernization is considered a critical component of enterprise transformation where enterprises are looking not merely to migrate legacy systems but also augment them with additional capabilities such as analytics, Internet of Things (IoT), and machine learning (ML) services in the cloud.

What are your top drivers for migrating SAP applications to the cloud?



AWS FOR SAP

Get started with AWS for SAP

AWS has been partnering with SAP in the cloud since 2008 and supporting customers' production SAP workloads since 2012. Today, many of SAP's cloud solutions run on AWS, and SAP customers are using AWS to migrate and modernize virtually every kind of SAP workload. Whether you want to improve reliability and security, simplify operations, and reduce costs for existing SAP applications or migrate to RISE with SAP or S/4HANA native, AWS can help eliminate the risk of your migration.

Differentiate your customer experience, accelerate growth, enhance sustainability, and improve business process efficiency by combining your SAP applications and data with any of the more than 200 services available from AWS. Drive new insights and improve real-time visibility across your business with a unified view of data from your ERP and other enterprise systems.

From data movement, data storage, data lakes, big data analytics, and log analytics to streaming analytics, business intelligence (BI), and ML, AWS offers purpose-built services that provide the best price performance and scalability.

Start your journey with purpose-built SAP services, such as **AWS Launch Wizard**, which allows you to deploy a production-ready SAP system on AWS in hours compared to weeks or months. Get expert help from a certified **AWS SAP Competency Partner** or the AWS Professional Services SAP team to help guide your implementation, migration, and innovation.

No matter what SAP system you run or strategy you'd like to adopt, AWS and our Certified Partners have the experience, tooling, methods, and best practices to streamline your migration and step up your transformation.

AWS for VMware

VMware tools have become a critical component of most enterprise IT departments, starting from the workstation and evolving to virtual machines as applications move to the server.

As the industry leader in server virtualization, VMware expanded to support more complex virtualized environments—starting with servers and storage and moving to management, DevOps, networking, security, and the development layer—which are all delivered through the provider's software-defined data center (SDDC) stack, the foundation of VMware Cloud.

VMware built the VMware Cloud platform to allow developers to build and deploy applications to any cloud by providing a consistent foundation for the integration of VMware vSphere, VMware vSAN, and VMware NSX virtualization technologies (among others) to simplify cloud migration for VMware users.

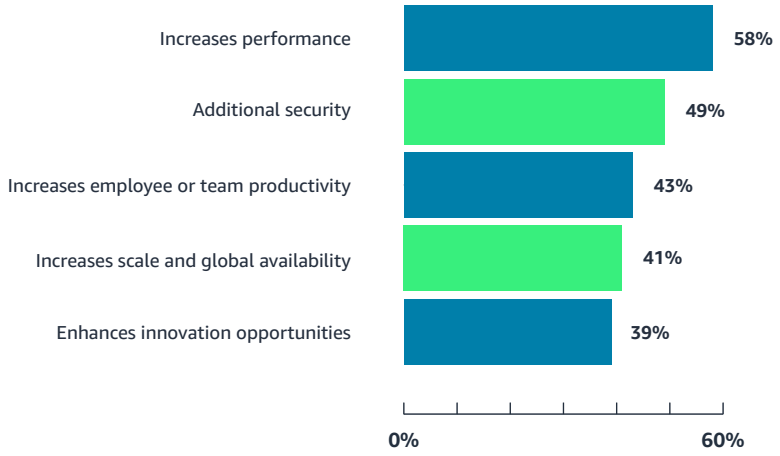


Key findings: VMware workloads

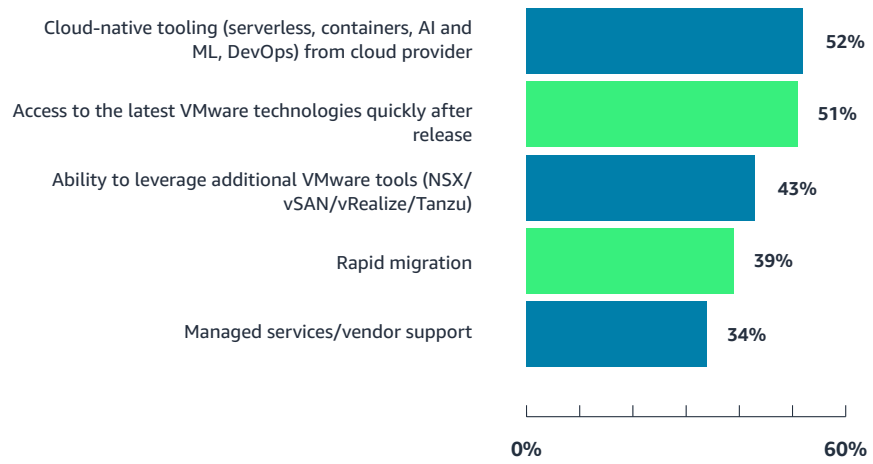
Enterprises identify key benefits of cloud infrastructure (scale, performance, security) and cloud services (innovation, productivity) as key growth drivers for migrating VMware-based applications to the cloud. This is aligned with other data showing that more than 50 percent of enterprises cite infrastructure refresh and data center consolidation as the primary drivers for moving their VMware workloads to the cloud.

Supportive of the interest in greater innovation (shown above), enterprises are eager to leverage cloud-native services and additional functionality to extend and augment VMware application development.

How does migration of VMware-based applications to the cloud directly contribute to growth?



Which factors are most important to you in choosing a cloud provider for your VMware stack and applications?



AWS FOR VMWARE

Get started with AWS for VMware

AWS and VMware have partnered to deliver VMware Cloud on AWS, our jointly engineered solution designed with speed and security in mind. Enterprises with investments in VMware virtualized applications and infrastructure can accelerate business transformation goals with a managed service that combines compute, network, and storage capabilities in a service that is maintained, supported, and operated by the creators of the software, VMware, and the leading public cloud provider, AWS.

Migrating to VMware Cloud on AWS provides access to more than 200 AWS services and VMware solutions to modernize your applications, unlock the value of your data, and accelerate innovation with industry-leading cloud technologies, including containers, serverless, specialized compute instances, and ML. Start your VMware Cloud on AWS journey with an **AWS Optimization and Licensing Assessment** (AWS OLA), engage our VMware specialists to assist with migration and modernization planning, or start your **14-day free trial** anytime.



AWS for Oracle applications

As a major provider of business applications, Oracle offers solutions, including E-Business Suite, PeopleSoft, JD Edwards, Hyperion, and Siebel. These are core components of a modern enterprise. As on-premises applications, however, they can be storage intensive, database intensive, and memory and CPU intensive, requiring continual and costly scaling of infrastructure to meet performance requirements.

Due to the on-premises nature of these applications, they lack the flexibility and scalability to adopt and respond to new and ever-changing business environments and trends. Innovations in the cloud, spanning data lakes, ML, and artificial intelligence (AI), are providing users with access to business insights automation of once-manual processes. In addition, the creation of purpose-built cloud databases and specialized compute instances is driving many enterprises to explore cloud infrastructure to meet that need.

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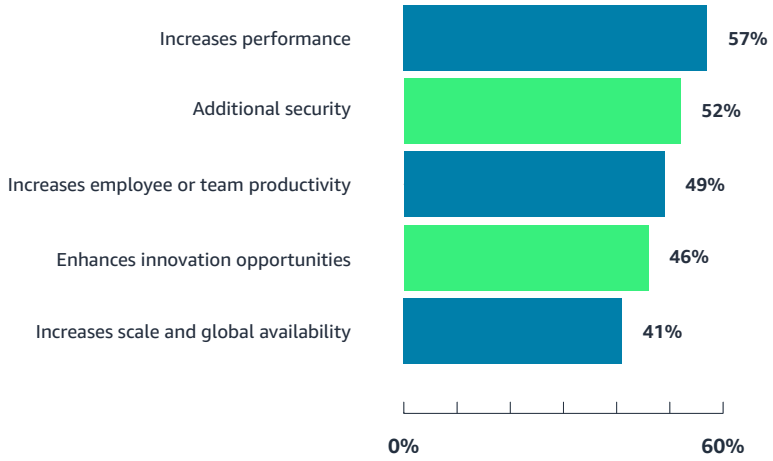


Key findings: Oracle workloads

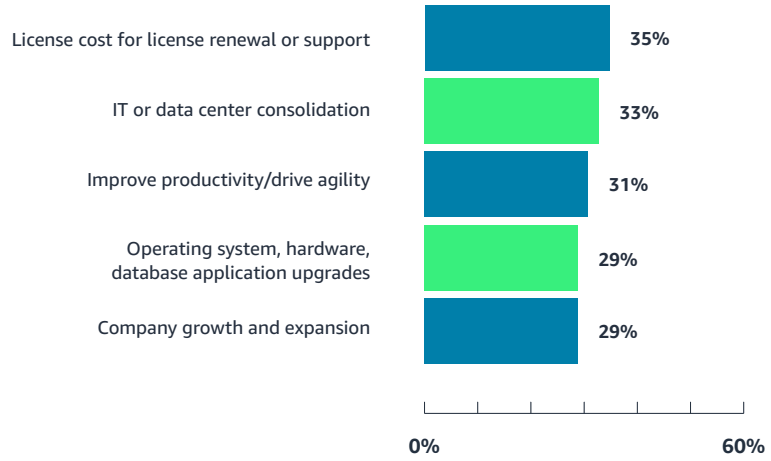
In addition to leveraging the cloud for scale, performance, and security, enterprise buyers are looking to the cloud to drive productivity improvements from improved accessibility, near real-time processing, and automation of routine tasks.

As with other enterprise workloads covered in this eBook, cost is a major factor in the cloud migration decision. In the case of Oracle applications, cost can be represented in licensing and support and associated infrastructure.

How does migration of Oracle ERP applications to the cloud directly contribute to growth?



What are your top drivers for migrating Oracle ERP applications to the cloud?



AWS FOR ORACLE APPLICATIONS

Get started with AWS for Oracle applications

Transform your Oracle applications by migrating and modernizing with AWS. Leverage more than 10 years of experience with mission-critical Oracle applications in the cloud. Learn how E-Business Suite, PeopleSoft, JD Edwards, Hyperion, and Siebel can run in the cloud to reduce total cost of ownership (TCO), increase efficiency, drive innovation, and transform customer experiences.

Evolve your Oracle applications, data, and operations with the world's most secure, reliable, and scalable cloud infrastructure. Leverage more than 200 cloud-native services, including leading AI and ML capabilities, to transform your business. Put the proven migration and modernization methodologies, services, and programs from AWS and our partners to work for your business. **Evaluate your Oracle licensing** in AWS or contact a certified **AWS Oracle Competency Partner** today.



AWS for mainframe workloads

Across a variety of industries, today's IT teams want to modernize their mainframe-based applications to take advantage of the cloud and leverage the years of business knowledge built into their systems for innovation. Augmenting, refactoring, or re-platforming mainframe applications to the AWS Cloud allows organizations to address the associated skill shortage and cost challenges. Customers benefit from the elasticity, scalability, performance, and reliability of AWS infrastructure and the flexibility to experiment with innovating with cloud services.

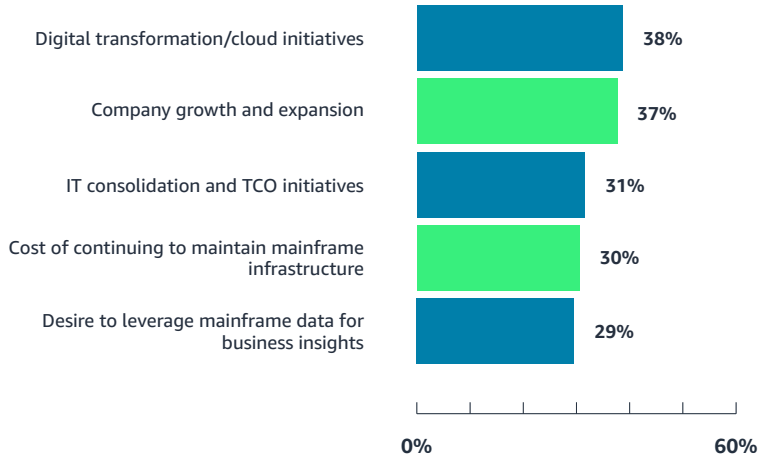


Key findings: Mainframe workloads

Enterprises are increasingly looking to the cloud as a way of modernizing and migrating mainframe systems and applications.

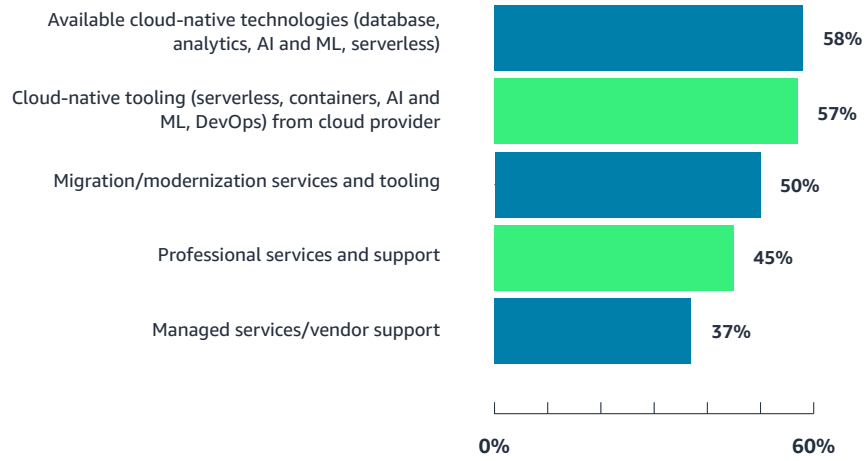
More than 40 percent of survey respondents are already migrating or planning to migrate mainframe applications this year. In addition to inclusion in digital transformation and consolidation strategies, the rising costs of maintaining mainframe workloads and leveraging mainframe data were among the top five drivers for enterprise teams migrating that data to the cloud.

What are your top drivers for migrating mainframe applications and data to the cloud?



As enterprises look to modernize mainframe applications, they are looking beyond just application migration and modernization. Many infrastructure teams are heavily considering the availability of cloud-native tooling and technologies as a means to drive additional agility and business insights.

Which factors are most important to you in choosing a cloud provider for mainframe modernization?



Get started with AWS for mainframe workloads

AWS Mainframe Modernization is a cloud-native, fully managed service that provides unified access to mainframe application modernization tools, middleware, and underlying infrastructure through the AWS native console. Using the AWS Mainframe Modernization service and associated methodologies and assessments, you can discover mainframe legacy applications, assess migration readiness, and plan modernization projects.

Once the planning is complete, you can choose from a variety of modernization patterns, including refactoring, re-platforming, and data augmentation to modernize mainframe applications with the AWS Cloud or leverage data from mainframe-based applications to run modern services for reporting, web/mobile integrations, analytics, AI, and ML in the AWS Cloud. The AWS Mainframe Modernization service lightens the effort of mainframe application modernization using automation and frees-up your teams to focus on agile development and rapid innovation to deliver higher business value. Get started with the **AWS Migration Acceleration Program (AWS MAP) for Mainframe**.

PROVEN BENEFITS

Why migrate and modernize with AWS?

From lifting and shifting workloads to moving entire data centers, AWS provides the organizational, operational, and technical capabilities you need for a successful cloud migration. And with the deepest set of modernization services, programs, and partners specializing in Microsoft, Oracle, SAP, VMware and mainframe applications and systems, you can realize the business value of AWS quickly.

Cost savings

Shift IT from a cost center to an innovation engine

Now more than ever, our customers are under pressure to cut costs. IT infrastructure and data center costs are often at the center of this discussion, especially as they are faced with a need for hardware upgrades and continual capacity and infrastructure additions. Another critical cost factor is enterprise software licensing in the form of ongoing license renewals and vendor pressure to move to the cloud.

A recent study found that software licensing costs can be as much as three times the cost of all other costs in the cloud combined (compute, network, and storage, for example) and that AWS can help customers reduce license needs for Windows Server by as much as 77 percent.[1]

AWS delivers on the promise of helping customers break free from the costly upgrade-and-refresh cycle by helping customers assess, optimize, and rightsize licensing and infrastructure for these applications. Use only what you need, replace licenses with cloud-based applications, or switch to open-source alternatives when appropriate.

"Software licensing costs can be as much as 3x the cost of all other costs in the cloud combined."

Enterprise Strategy Group (ESG), 2022[1]

[1] Strechay, R., "Reduce Licensing and Infrastructure Costs while on the Move to the AWS Cloud," ESG, April 2022

Data transformation

Your data is your differentiator

What you do with applications and data in the cloud is where digital transformation really begins. When it comes to legacy enterprise applications, it's the employee, customer, and transactional data these systems process and store that should be the foundation of your cloud-enabled efforts to drive innovation and differentiation. You likely have years (perhaps even decades) of data tied up in ERP applications, CRM systems, and mainframes that may be processing millions of records or transactions.

AWS has the experience, programs, partners, and specialized services to not only migrate applications and databases to the cloud but also the broadest set of analytics, AI, ML, and scalable database options to help you leverage that business data to drive new and differentiated experiences for employees and customers. Use industry-leading data and analytics tools from AWS to leverage your data in a way that can drive real differentiation for your business.

Agility and productivity

With AWS, innovation is built in

Moving enterprise applications and data to the AWS Cloud opens opportunities for you to rethink how your teams build and deliver software. A move to AWS should include planning for transitioning people skills to modern cloud operations and agile software development.

Re-platforming and refactoring applications is an opportunity to drive change and efficiencies in your business. Simply stated, modernization and innovation start with migration. Working with AWS to lift and shift apps, data, and infrastructure today presents new opportunities to help you modernize tomorrow.

As you get comfortable with cloud operating models, you will gain the confidence to re-platform and refactor select applications to experiment with containers, Kubernetes, and serverless systems. You will also gain the ability to leverage more open-source technologies while you shift operating models away from waterfall processes to agile development. AWS is innovating faster than any other cloud provider, and each of the hundreds of cloud-native services is there when you're ready to modernize your development, operations, applications…and, ultimately, your business.

Scale and security

The last upgrade you will ever need

Enterprises are increasingly choosing AWS to host their cloud-based infrastructure. In the process, they are realizing increased performance, security, reliability, and scale wherever they go. Mission-critical enterprise applications need infrastructure that can meet and exceed the demands your business will put on them. The AWS Global Cloud Infrastructure is the most secure, extensive, and reliable cloud platform, offering more than 200 fully featured services from data centers globally.

Whether you need to deploy your application workloads across the globe in a single click or you want to build and scale specific applications closer to your end users with single-digit millisecond latency, AWS provides the cloud infrastructure where and when you need it. And the AWS Cloud is backed by a deep set of security tools, with compliance and governance services and features and support for 98 security standards and compliance certifications. All 117 AWS services that store customer data offer the ability to encrypt that data. Give your applications the scalable, secure infrastructure they need to drive the business outcomes you want.

BETTER OUTCOMES START HERE

Not all clouds are created equal

AWS has more than a decade of direct customer experience with thousands of the world's largest enterprises migrating and modernizing Microsoft, Oracle, SAP, VMware and mainframe workloads. This experience has helped us design and hone our tools, guidance, services, and programs to simplify digital transformation for our customers.

Get started today by choosing from our free industry-leading **resources**, **AWS Professional Services**, and more than 170 certified **AWS Migration Competency Partners** to guide your migration and modernization journey. Our approach to migrating and modernizing enterprise workloads is driving meaningful outcomes for our customers across cost savings, staff productivity, operational resilience, and business agility.

Get started on your enterprise workload migration and modernization initiatives today with AWS.

Learn more ›

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